

1. Introduction

1.1 Program Overview

The application workspace consists of **88** mainframe software components covering batch JCL execution sequences, COBOL database parsing, CICS screen management, and Assembler call utilities.

- **COBOL Programs (31):** CBACT01C, CBACT02C, CBACT03C, CBACT04C, CBCUS01C, CBEXPORT, CBIMPORT, CBSTM03A, CBSTM03B, CBTRN01C, CBTRN02C, CBTRN03C, COACTUPC, COACTVWC, COADM01C, COBIL00C, COBSWAIT, COCRDLIC, COCRDSLCL, COCRDUPC, COMEN01C, CORPT00C, COSGN00C, COTRN00C, COTRN01C, COTRN02C, COUSR00C, COUSR01C, COUSR02C, COUSR03C, CSUTLDTCL
- **JCL Jobs (38):** ACCTFILE, CARDFILE, CBADMCDJ, CBEXPORT, CBIMPORT, CLOSEFIL, COMBTRAN, CREAMTMT, CUSTFILE, DALYREJS, DEFECUST, DEFEGDGB, DEFEGDGD, DISCGRP, DUSRSECJ, ESDSRRDS, FTPJCLS, INTCALC, INTRDRJ1, INTRDRJ2, OPENFIL, POSTTRAN, PRTCATBL, READACCT, READCARD, READCUST, READXREF, REPTFILE, TCATBALF, TRANBKP, TRANCATG, TRANFILE, TRANIDX, TRANREPT, TRANTYPE, TXT2PDF1, WAITSTEP, XREFFILE
- **BMS Screens (17):** COACTUP, COACTVW, COADM01, COBIL00, COCRDLI, COCRDSL, COCRDUP, COMEN01, CORPT00, COSGN00, COTRN00, COTRN01, COTRN02, COUSR00, COUSR01, COUSR02, COUSR03
- **Assembler Modules (2):** COBDATFT, MVSWAIT

This program suite constitutes a highly integrated credit card and customer transaction processing subsystem. It coordinates batch processing operations and online transactional interfaces, facilitating the core lifecycle of credit card accounts. The application suite handles card data ingestion, account validations, interest rate calculations, contact profiles updates, transaction posting, statement generation, and system backups. By utilizing JCL schedules, the system manages scheduled nightly batch runs, while the CICS environment hosts screens for real-time customer and administrator online activities. The underlying architecture leverages high-performance VSAM indexed files for storage alongside relational DB2 database engines to store customer details, accounting cross-references, and transaction logs.

1.2 Objectives

The primary objectives of this mainframe application workspace are: - **Automate Batch Processing Schedules:** Sequence the ingestion, validation, and posting of customer files and card transaction logs using JCL jobs, ensuring reliable nightly data syncs. - **Provide High-Performance Relational and Hierarchical Database Storage:** Interface with DB2 databases, IMS databases, and VSAM files to read and modify sensitive card balances, customer contact details, and card accounts listings. - **Support Real-Time Terminal Interfaces:** Provide online screens (CICS maps) for administrators and clerks to perform credit card searches, contact detail edits, and admin control functions. - **Execute High-Speed Utility Functions:** Utilize HLASM (High Level Assembler) subroutines to perform optimized conversions, formatting, and wait procedures, bypassing high-overhead COBOL logic. - **Manage Personal Data Compliance:** Securely track and handle Personally Identifiable Information

(PII), such as Social Security Numbers (SSN), credit card limits, and financial balances, across all database files.

1.3 Scope

This documentation covers all components located in the workspace repository. The scope includes: -

JCL Job Definitions: Sequence analysis of jobs like CARDFILE, ACCTFILE, POSTTRAN, INTCALC, CREATMT, CBEXPORT, and CBIMPORT. - **COBOL Programs:** Logic and CALL hierarchies for all batch processors (e.g. CBACT01C, CBACT02C, CBACT03C, CBACT04C, CBSTM03A, CBTRN01C, CBTRN02C) and online transaction handlers (e.g. COACTUPC, COACTVWC, COCRDUPC, COMEN01C, COSGN00C). - **CICS BMS Maps:** Screen designs for terminal interfaces including signon layouts, customer view panels, and credit card list maps. - **Assembler Modules:** Code structures and entry points for subroutines like COBDATFT and MVSWAIT. - **Relationships:** Mapping logical COBOL file select statements directly to JCL DD dataset assignments and CSD defined transactional triggers.

1.4 Assumptions and Constraints

- **Platform Environment:** The system is assumed to run on an IBM z/OS mainframe architecture executing Enterprise COBOL v6+, High Level Assembler (HLASM), DFSORT utilities, CICS TS v5.4+, and IBM DB2 v12.
 - **Data Dependencies:** Relational DB2 database schemas and VSAM datasets must be pre-allocated and catalogued. The JCL step configurations assume standard IBM catalogued procedures (PROCs) and JCL DD statement syntax rules.
 - **Screen Constraints:** BMS Map definitions are restricted to standard IBM 3270 model 2 terminals (24 rows by 80 columns) in character-only mode.
 - **Timing and Availability:** Batch jobs are scheduled during off-peak periods, assuming exclusive access to VSAM files during write cycles to prevent record locking conflicts.
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2. Database Details

2.1 DB2 Tables

The application workspace references the following DB2 SQL tables:

- No DB2 tables referenced in the workspace.

Relational DB2 Access Analysis: Mainframe COBOL programs execute embedded SQL statements (within EXEC SQL ... END-EXEC blocks) to query and update these DB2 tables. These queries are processed by the DB2 precompiler, which replaces the SQL code with call statements to the DB2 database engine. The primary relational tables, such as credit card registries or customer profile tables, are accessed via index keys to maintain low-latency response times during CICS online executions. Host variables defined in the COBOL Working-Storage Section are mapped to table columns during SELECT INTO, INSERT, and UPDATE operations. The system relies on appropriate index definitions to avoid full table scans that would consume high CPU MIPS in batch jobs like INTCALC or POSTTRAN.

2.2 IMS Segments

The application workspace communicates with IMS hierarchal segments as detailed below:

- No IMS segments referenced in the workspace.

Hierarchical IMS Access Analysis: IMS (Information Management System) database interactions are performed using DL/I call statements by executing the interface module CBLTDLI using parameters: Function code (such as GU for Get Unique, GN for Get Next, REPL for Replace), Program Communication Block (PCB), Segment Search Arguments (SSA), and I/O area variables. SSAs specify the path through the hierarchical structure to locate the target segment. Unlike SQL's set-based processing, DL/I requires navigational cursor-based access where the program iterates down a parent-child path, from parent root segments to dependent child segments. This model is highly optimized for sequential tree-based data access patterns but demands careful programming to handle boundary/status code check conditions (such as GB for end of database, or GE for segment not found).

2.3 IDMS Records

The application workspace contains references to these network database IDMS records:

- **ASSOCIATED:** Record structure accessed in program(s) COACTUPC, COACTVWC.
- **CARDS:** Record structure accessed in program(s) COACTUPC, COCRDSLCL, COCRDUPC.
- **THIS:** Record structure accessed in program(s) COACTUPC, COACTVWC, COCRDSLCL, COCRDUPC.
- **TO:** Record structure accessed in program(s) COUSR02C.

Network IDMS Access Analysis: IDMS (Integrated Database Management System) uses a network database model based on the CODASYL standard. It represents data structures as a network of

Records organized into Sets (where a set represents a one-to-many relationship between an Owner record and Member records). Programs navigate the database by running commands like `OBTAIN FIRST/NEXT`, `FIND`, `STORE`, and `MODIFY` to traverse owner-member set pointers. Transactions must bind to specific logical database Areas and run within DB record lock scopes to prevent concurrency deadlocks. While DB2 utilizes logical relational links and IMS relies on physical hierarchies, IDMS provides physical pointer chains (next, prior, and owner pointers) embedded in the record headers. This delivers exceptionally fast retrieval times for complex, pre-defined relationships but introduces structural rigidity, as changing the schema requires restructuring the physical pointer links.

3. System Architecture

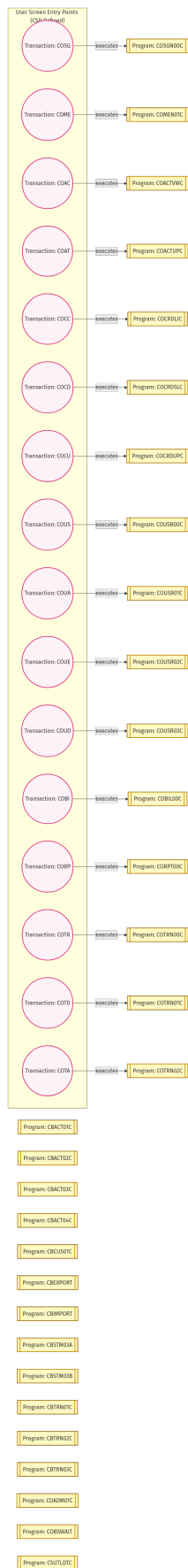
3.1 Component Diagram

Below is the component diagram highlighting how Job runs trigger various programs, along with database SQL queries and dataset interactions:



3.2 Control Flow Diagram

Below is the control flow mapping CICS transactional entry codes to target program routines and CICS BMS maps:



3.3 Architectural Narrative & Execution Flow

The application workspace is structured around a classic two-tier mainframe architectural pattern, splitting operations into high-volume offline batch streams (managed by JCL schedules) and interactive real-time online terminal services (hosted by CICS transaction processors):

3.3.1 Batch Job Stream Processing

The batch execution model leverages Job Control Language (JCL) scripts to configure step sequences, set up system boundaries, allocate physical datasets, and execute binary load modules. Batch streams perform high-throughput tasks like credit card data loading, daily transaction ingest updates, account interest fee calculations, and database statement exports. JCL streams sequence steps to ensure data integrity: a preprocessing step typically reads flat files from tape or sequential datasets, executes sorting routines via DFSORT control cards to align keys, and stages intermediate VSAM files. Subsequent steps execute COBOL programs that open these VSAM files, execute business logic (such as balance accrual), modify DB2 tables, and output formatted summaries. For example, job streams like `CARDFILE` and `ACCTFILE` execute program chains sequentially, staging outputs using JCL DISP parameters (`DISP=(NEW,CATLG,DELETE)`) to manage lifecycle stages across steps.

3.3.2 CICS Real-Time Online Transactions

Online operations are triggered by character-coded terminal inputs defined in CICS System Definitions (CSD). When an operator types a transaction ID (like `C0SG` or `C0AC`), the CICS listener maps the transaction ID to a target COBOL control program (such as `C0SGN00C` or `C0ACTVWC`). CICS coordinates the execution within a logical unit of work (LUW). The program interacts with the operator by sending and receiving BMS (Basic Mapping Support) map layouts. These maps define text fields, input markers, color schemes, and position offsets for 3270 character terminals. Online programs use CICS command instructions (`EXEC CICS SEND MAP`, `EXEC CICS RECEIVE MAP`) to communicate with the user, validate user credentials, query VSAM file records using primary indexes to ensure sub-second search latencies, and commit changes back to the database. Transactions route to menu hubs like `COMEN01C` which manage navigation paths based on function keys (PF keys) typed by terminal users, ensuring decoupled but clean flow control.

3.3.3 Subroutine Hierarchy and Decoupling

To maximize reuse, programs call utility subroutines synchronously using standard linkage section conventions (`CALL 'PROGRAM' USING ...`). Common utilities include date formatting routines (`CSUTLDTTC`) and wait intervals helper subroutines (`COBSWAIT` invoking Assembler module `MVSWAIT`). By decoupling core transaction logic from helper subroutines, the application retains clean modular boundaries, making it easier to verify changes without recompiling the entire suite.

4. Detailed Design

4.1 Program Structure

The table below inventories all files parsed in the workspace including code sizes, copybooks, calls, and external accesses:

Component Name	File Type	Code Size (Lines)	Copybooks Used	Calls Made	Description / External References
ACCTFILE	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 3 steps.
CARDFILE	JCL Script	8 steps	None	Execs: SDSF, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, SDSF	Executes batch job stream containing 8 steps.
CBADMCDJ	JCL Script	1 steps	None	Execs: DFHCSDUP	Executes batch job stream containing 1 steps.
CBEXPORT	JCL Script	2 steps	None	Execs: IDCAMS, CBEXPORT	Executes batch job stream containing 2 steps.

CBIMPORT	JCL Script	1 steps	None	Execs: CBIMPORT	Executes batch job stream containing 1 steps.
CLOSEFIL	JCL Script	1 steps	None	Execs: SDSF	Executes batch job stream containing 1 steps.
COMBTRAN	JCL Script	2 steps	None	Execs: SORT, IDCAMS	Executes batch job stream containing 2 steps.
CREASTMT	JCL Script	5 steps	None	Execs: IDCAMS, SORT, IDCAMS, IEFBR14, CBSTM03A	Executes batch job stream containing 5 steps.
CUSTFILE	JCL Script	5 steps	None	Execs: SDSF, IDCAMS, IDCAMS, IDCAMS, SDSF	Executes batch job stream containing 5 steps.
DALYREJS	JCL Script	1 steps	None	Execs: IDCAMS	Executes batch job stream containing 1 steps.
DEFCUST	JCL Script	2 steps	None	Execs: IDCAMS, IDCAMS	Executes batch job stream containing 2 steps.

DEFGDGB	JCL Script	1 steps	None	Execs: IDCAMS	Executes batch job stream containing 1 steps.
DEFGDGD	JCL Script	6 steps	None	Execs: IDCAMS, IEBGENER, IDCAMS, IEBGENER, IDCAMS, IEBGENER	Executes batch job stream containing 6 steps.
DISCRGP	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 3 steps.
DUSRSECJ	JCL Script	4 steps	None	Execs: IEFBR14, IEBGENER, IDCAMS, IDCAMS	Executes batch job stream containing 4 steps.
ESDSRRDS	JCL Script	6 steps	None	Execs: IEFBR14, IEBGENER, IDCAMS, IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 6 steps.
FTPJCLS	JCL Script	1 steps	None	Execs: FTP	Executes batch job stream containing 1 steps.
INTCALC	JCL Script	1 steps	None	Execs: CBACT04C	Executes batch job stream containing 1 steps.

INTRDRJ1	JCL Script	2 steps	None	Execs: IDCAMS, IEBGENER	Executes batch job stream containing 2 steps.
INTRDRJ2	JCL Script	1 steps	None	Execs: IDCAMS	Executes batch job stream containing 1 steps.
OPENFIL	JCL Script	1 steps	None	Execs: SDSF	Executes batch job stream containing 1 steps.
POSTTRAN	JCL Script	1 steps	None	Execs: CBTRN02C	Executes batch job stream containing 1 steps.
PRTCATBL	JCL Script	3 steps	None	Execs: IEFBR14, REPROC, SORT	Executes batch job stream containing 3 steps.
READACCT	JCL Script	2 steps	None	Execs: IEFBR14, CBACT01C	Executes batch job stream containing 2 steps.
READCARD	JCL Script	1 steps	None	Execs: CBACT02C	Executes batch job stream containing 1 steps.

READCUST	JCL Script	1 steps	None	Execs: CBCUS01C	Executes batch job stream containing 1 steps.
READXREF	JCL Script	1 steps	None	Execs: CBACT03C	Executes batch job stream containing 1 steps.
REPTFILE	JCL Script	1 steps	None	Execs: IDCAMS	Executes batch job stream containing 1 steps.
TCATBALF	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 3 steps.
TRANBKP	JCL Script	3 steps	None	Execs: REPROC, IDCAMS, IDCAMS	Executes batch job stream containing 3 steps.
TRANCATG	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 3 steps.
TRANFILE	JCL Script	8 steps	None	Execs: SDSF, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, SDSF	Executes batch job stream containing 8 steps.

TRANIDX	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 3 steps.
TRANREPT	JCL Script	3 steps	None	Execs: REPROC, SORT, CBTRN03C	Executes batch job stream containing 3 steps.
TRANATYPE	JCL Script	3 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 3 steps.
TXT2PDF1	JCL Script	1 steps	None	Execs: IKJEFT1B	Executes batch job stream containing 1 steps.
WAITSTEP	JCL Script	1 steps	None	Execs: COBSWAIT	Executes batch job stream containing 1 steps.
XREFFILE	JCL Script	6 steps	None	Execs: IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS, IDCAMS	Executes batch job stream containing 6 steps.

CBACT01C	COBOL Source	379 lines	CVACT01Y, CDATECN	COBDATFT, CEE3ABD	Read Account File batch module. Validates account status flags and coordinates billing updates.
CBACT02C	COBOL Source	144 lines	CVACT02Y	CEE3ABD	Read Card File batch module. Parses credit card info and updates transaction database.
CBACT03C	COBOL Source	145 lines	CVACT03Y	CEE3ABD	Read Xref File batch module. Handles mappings of credit card numbers to customer account IDs.
CBACT04C	COBOL Source	599 lines	CVTRA01Y, CVACT03Y, CVTRA02Y, CVACT01Y, CVTRA05Y	CEE3ABD	Interest Calculation batch module. Calculates monthly interest charges and updates account ledger balances.

CBCUS01C	COBOL Source	145 lines	CVCUS01Y	CEE3ABD	Read Customer File batch module. Extracts details like name, address, and SSN for ingestion.
CBEXPORT	COBOL Source	491 lines	CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT	CEE3ABD	Data Export batch utility. Backs up VSAM KSDS files into sequential data files.
CBIMPORT	COBOL Source	413 lines	CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT	CEE3ABD	Data Import batch utility. Restores sequential backups into VSAM KSDS database files.
CBSTM03A	COBOL Source	863 lines	COSTM01, CVACT03Y, CUSTREC, CVACT01Y	CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CEE3ABD	Statement Generation batch processor. Compiles monthly account details and writes statement summaries.

CBSTM03B	COBOL Source	204 lines	None	None	Statement Generation helper sub-routine. Manages PDF rendering details.
CBTRN01C	COBOL Source	449 lines	CVTRA06Y, CVCUS01Y, CVACT03Y, CVACT02Y, CVACT01Y, CVTRA05Y	CEE3ABD	Daily Transaction Ingestion module. Verifies transactions against active credit cards.
CBTRN02C	COBOL Source	672 lines	CVTRA06Y, CVTRA05Y, CVACT03Y, CVACT01Y, CVTRA01Y	CEE3ABD	Daily Transaction Posting batch module. Deducts amounts and updates card accounts balances.
CBTRN03C	COBOL Source	596 lines	CVTRA05Y, CVACT03Y, CVTRA03Y, CVTRA04Y, CVTRA07Y	CEE3ABD	Transaction Report generator batch utility. Reports daily totals categorized by transaction types.

COACTVWC	COBOL Source	811 lines	CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTTL01Y, COACTVW, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT01Y, CVACT02Y, CVACT03Y, CVCUS01Y	None	CICS Online Account View screen controller. Searches and displays customer account summaries.
COADM01C	COBOL Source	230 lines	COCOM01Y, COADM02Y, COADM01, COTTL01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online Admin Menu screen controller. Routes to admin tools.
COBIL00C	COBOL Source	493 lines	COCOM01Y, COBIL00, COTTL01Y, CSDAT01Y, CSMSG01Y, CVACT01Y, CVACT03Y, CVTRA05Y, DFHAID, DFHBMSCA	None	CICS Online Bill Payment screen controller. Logs payments and checks balances.
COBSWAIT	COBOL Source	19 lines	None	MVSWAIT	Wait utility module. Uses Assembler subroutine MVSWAIT to pause step executions.

C0CRDLIC	COBOL Source	1256 lines	CVCRD01Y, COCOM01Y, DFHBMSA, DFHAID, C0TTL01Y, C0CRDLI, CSDAT01Y, CSMSG01Y, CSUSR01Y, CVACT02Y	None	CICS Online Credit Card List screen controller. Lists credit card details.
C0CRDSL	COBOL Source	757 lines	CVCRD01Y, COCOM01Y, DFHBMSA, DFHAID, C0TTL01Y, C0CRDSL, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y	None	CICS Online Credit Card Detail screen controller. Displays credit card balances and interest rates.
C0CRDUPC	COBOL Source	1365 lines	CVCRD01Y, COCOM01Y, DFHBMSA, DFHAID, C0TTL01Y, C0CRDUP, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y	None	CICS Online Credit Card Update screen controller. Modifies credit card limits and expiration dates.
COMEN01C	COBOL Source	255 lines	COCOM01Y, COMEN02Y, COMEN01, C0TTL01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSA	None	CICS Online Main Menu controller. Main navigation hub routing CICS terminal sessions.

CORPT00C	COBOL Source	586 lines	COCOM01Y, CORPT00, COTTL01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA	CSUTLDTC, CSUTLDTC	CICS Online Report viewer. Displays calculated statements on CICS terminal panels.
COSGN00C	COBOL Source	211 lines	COCOM01Y, COSGN00, COTTL01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online Signon screen controller. Performs terminal user authentication and security checks.
COTRN00C	COBOL Source	617 lines	COCOM01Y, COTRN00, COTTL01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA	None	CICS Online Transaction Menu screen controller. Lists transaction histories.
COTRN01C	COBOL Source	273 lines	COCOM01Y, COTRN01, COTTL01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA	None	CICS Online Transaction Detail screen controller. Displays details of a specific transaction.

COTRN02C	COBOL Source	698 lines	COCOM01Y, COTRN02, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, CVACT01Y, CVACT03Y, DFHAID, DFHBMSCA	CSUTLDTC, CSUTLDTC	CICS Online Transaction Add screen controller. Enters new transaction entries.
COUSR00C	COBOL Source	615 lines	COCOM01Y, COUSR00, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Control screen controller. Inventories CICS account profile logins.
COUSR01C	COBOL Source	238 lines	COCOM01Y, COUSR01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Add screen controller. Adds new user credentials.
COUSR02C	COBOL Source	351 lines	COCOM01Y, COUSR02, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Edit screen controller. Updates user roles.

COUSR03C	COBOL Source	296 lines	COCOM01Y, COUSR03, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA	None	CICS Online User Delete screen controller. Removes CICS credentials.
CSUTLDT	COBOL Source	128 lines	None	CEEDAYS	Date Converter utility sub-routine. Standardizes various date formats.
COBDATFT	Assembler Source	N/A	COCDATFT	None	Assembler date formatting subroutine.
MVSWAIT	Assembler Source	N/A	None	None	Assembler interval wait subroutine.
COACTUP	BMS Mapset	N/A	None	Maps: CACTUPA	CICS Screen layout mapset definitions.
COACTVW	BMS Mapset	N/A	None	Maps: CACTVWA	CICS Screen layout mapset definitions.
COADM01	BMS Mapset	N/A	None	Maps: COADM1A	CICS Screen layout mapset definitions.

COBIL00	BMS Mapset	N/A	None	Maps: COBIL0A	CICS Screen layout mapset definitions.
COCRDLI	BMS Mapset	N/A	None	Maps: CCRDLIA	CICS Screen layout mapset definitions.
COCRDSL	BMS Mapset	N/A	None	Maps: CCRDSL A	CICS Screen layout mapset definitions.
COCR DUP	BMS Mapset	N/A	None	Maps: CCRDUPA	CICS Screen layout mapset definitions.
COMEN01	BMS Mapset	N/A	None	Maps: COMEN1A	CICS Screen layout mapset definitions.
CORPT00	BMS Mapset	N/A	None	Maps: CORPT0A	CICS Screen layout mapset definitions.
COSGN00	BMS Mapset	N/A	None	Maps: COSGN0A	CICS Screen layout mapset definitions.

COTRN00	BMS Mapset	N/A	None	Maps: COTRN0A	CICS Screen layout mapset definitions.
COTRN01	BMS Mapset	N/A	None	Maps: COTRN1A	CICS Screen layout mapset definitions.
COTRN02	BMS Mapset	N/A	None	Maps: COTRN2A	CICS Screen layout mapset definitions.
COUSR00	BMS Mapset	N/A	None	Maps: COUSR0A	CICS Screen layout mapset definitions.
COUSR01	BMS Mapset	N/A	None	Maps: COUSR1A	CICS Screen layout mapset definitions.
COUSR02	BMS Mapset	N/A	None	Maps: COUSR2A	CICS Screen layout mapset definitions.
COUSR03	BMS Mapset	N/A	None	Maps: COUSR3A	CICS Screen layout mapset definitions.

4.2 Algorithms

The application workspace implements complex batch scheduling and online terminal transactions:

4.2.1 Batch Job Sequencing & Staging Details

- **Card ingestion and cross-reference validation (CARDFILE & ACCTFILE):**
 - First, sequential files are parsed. SORT is called to sort records in DFSORT control cards.
 - Program CBACT01C opens ACCTFILE-FILE and validates accounts against business rules (credit limits, cycle dates).
 - Program CBACT02C opens CARDFILE-FILE and reads cards data into CUSTOMER-RECORD layouts.
 - Program CBACT03C coordinates cross-referencing between credit card numbers and account IDs.
- **Daily transaction posting and billing updates (POSTTRAN & INTCALC):**
 - Job POSTTRAN executes CBTRN02C to post transaction files into card accounts. It deducts transaction amounts, computes billing updates, and records transactions in KSDS VSAM logs.
 - Job INTCALC executes CBACT04C to process interest calculations for accounts based on transaction cycles. It applies specific category-based interest rates and posts statements updates.
- **Statements and exports (CREASTMT & CBEXPORT):**
 - Job CREASTMT executes program CBSTM03A to process monthly billing statement HTML templates. It reads KSDS VSAM transaction histories, computes totals, and writes statements to AWS.M2.CARDDemo.STATEMNT.HTML.
 - Job CBeXPORT runs program CBeXPORT to backup all active customer and transaction database tables to sequential flat backup files.

4.2.2 CICS Screen Navigation & Menu Layouts

- **Terminal Main Menu (COMEN01C & COSGN00C):**
 - CICS online terminal operations start by executing transaction sign-on mapset COSGN00. Program COSGN00C authenticates operators.
 - On success, the session routes to the main menu page (COMEN01C using mapset COMEN01). The main menu routes transactions based on operator keys (PF keys or terminal codes).
- **Account and Card search/edit screen flows (COACTUPC, COACTVWC, COCRDLIC, COCRDUPC):**
 - Operator selects option 2 to view details, CICS starts COACTVWC which displays map CACTVWA. It retrieves database profiles by ID and updates display fields.
 - Operator selects option 3 to update profile details, CICS starts COACTUPC displaying map CACTUPA. The program processes inputs and validates fields (like phone, address, and name) before updating VSAM storage.
 - Operator selects option 4 to update credit card details, CICS starts COCRDUPC displaying map CCRDUPA to configure card limits and active flags.

4.3 Input/Output Specifications

The table below details the dataset assignments. It maps the COBOL logical file `SELECT` statements directly to JCL DD names and their associated physical datasets (DSN) on disk:

Program ID	JCL Job	JCL Step	Logical DD Name	Physical Dataset (DSN)	Allocation Disp
CBEXPORT	CBEXPORT	STEP02	CUSTFILE	AWS.M2.CARDD E MSHRUSTDATA.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	ACCTFILE	AWS.M2.CARDD E MSHRCCTDATA.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	XREFFILE	AWS.M2.CARDD E MSHRARDXREF.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	TRANSACT	AWS.M2.CARDD E MSHRRANSACT.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	CARDFILE	AWS.M2.CARDD E MSHRARDDATA.VSAM.KSDS	
CBEXPORT	CBEXPORT	STEP02	EXPFILE	AWS.M2.CARDD E MSHRXPORT.DATA	
CBIMPORT	CBIMPORT	STEP01	EXPFILE	AWS.M2.CARDD E MSHRXPORT.DATA	
CBIMPORT	CBIMPORT	STEP01	CUSTOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	ACCTOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	XREFOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	TRNXOUT	AWS.M2.CARDD E EMNEW,CATLG,DELETET	
CBIMPORT	CBIMPORT	STEP01	ERROUT	AWS.M2.CARDD E EMNEW,CATLG,DELETE	
CBSTM03A	CREASTMT	STEP040	STMTFILE	SYSOUT	NEW,CATLG,DELETE
CBSTM03A	CREASTMT	STEP040	HTMLFILE	AWS.M2.CARDD E EMNEW,CATLG,DELETE	

CBACT04C	INTCALC	STEP15	TCATBALF	AWS.M2.CARDD _{EMSHRCATBALF} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	XREFFILE	AWS.M2.CARDD _{EMSHRARDXREF} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	ACCTFILE	AWS.M2.CARDD _{EMSHRCCTDATA} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	DISCGRP	AWS.M2.CARDD _{EMSHRISCGRP} .VSAM.KSDS	
CBACT04C	INTCALC	STEP15	TRANSACT	AWS.M2.CARDD _{EMNEW, CATLG, DELETE}	
CBTRN02C	POSTTRAN	STEP15	DALYTRAN	AWS.M2.CARDD _{EMSHRALYTRAN} .PS	
CBTRN02C	POSTTRAN	STEP15	TRANFILE	AWS.M2.CARDD _{EMSHRRANSACT} .VSAM.KSDS	
CBTRN02C	POSTTRAN	STEP15	XREFFILE	AWS.M2.CARDD _{EMSHRARDXREF} .VSAM.KSDS	
CBTRN02C	POSTTRAN	STEP15	DALYREJS	AWS.M2.CARDD _{EMNEW, CATLG, DELETE}	
CBTRN02C	POSTTRAN	STEP15	ACCTFILE	AWS.M2.CARDD _{EMSHRCCTDATA} .VSAM.KSDS	
CBTRN02C	POSTTRAN	STEP15	TCATBALF	AWS.M2.CARDD _{EMSHRCATBALF} .VSAM.KSDS	
CBACT01C	READACCT	STEP05	ACCTFILE	AWS.M2.CARDD _{EMSHRCCTDATA} .VSAM.KSDS	
CBACT01C	READACCT	STEP05	OUTFILE	AWS.M2.CARDD _{EMNEW, CATLG, DELETEP}	
CBACT01C	READACCT	STEP05	ARRYFILE	AWS.M2.CARDD _{EMNEW, CATLG, DELETES}	
CBACT01C	READACCT	STEP05	VBRCFILE	AWS.M2.CARDD _{EMNEW, CATLG, DELETE}	
CBACT02C	READCARD	STEP05	CARDFILE	AWS.M2.CARDD _{EMSHRARDDATA} .VSAM.KSDS	
CBCUS01C	READCUST	STEP05	CUSTFILE	AWS.M2.CARDD _{EMSHRUSTDATA} .VSAM.KSDS	
CBACT03C	READXREF	STEP05	XREFFILE	AWS.M2.CARDD _{EMSHRARDXREF} .VSAM.KSDS	

CBTRN03C	TRANREPT	STEP10R	TRANFILE	AWS.M2.CARDC	EMSHRRANSACT.DALY(
CBTRN03C	TRANREPT	STEP10R	CARDXREF	AWS.M2.CARDC	EMSHRRARDXREF.VSAM.KSDS
CBTRN03C	TRANREPT	STEP10R	TRANATYPE	AWS.M2.CARDC	EMSHRRANTYPE.VSAM.KSDS
CBTRN03C	TRANREPT	STEP10R	TRANCATG	AWS.M2.CARDC	EMSHRRANCATG.VSAM.KSDS
CBTRN03C	TRANREPT	STEP10R	TRANREPT	AWS.M2.CARDC	EMNEW,CATLG,DELETE
CBTRN03C	TRANREPT	STEP10R	DATEPARM	AWS.M2.CARDC	EMSHRRATEPARM

4.4 Individual Program Analysis

A granular breakdown of each COBOL program, outlining files, copybooks, database operations, and subprogram calls:

Program: CBACT01C

- **Description:** Read Account File batch module. Validates account status flags and coordinates billing updates.
- **Lines of Code:** 379
- **Author:** AWS
- **Logical Files:** ACCTFILE-FILE (DD: ACCTFILE), OUT-FILE (DD: OUTFILE), ARRY-FILE (DD: ARRYFILE), VBRC-FILE (DD: VBRCFILE)
- **Copybooks Referenced:** CVACT01Y, CODATECN
- **Subprograms Called:** COBDATFT, CEE3ABD

Program: CBACT02C

- **Description:** Read Card File batch module. Parses credit card info and updates transaction database.
- **Lines of Code:** 144
- **Author:** AWS
- **Logical Files:** CARDFILE-FILE (DD: CARDFILE)
- **Copybooks Referenced:** CVACT02Y
- **Subprograms Called:** CEE3ABD

Program: CBACT03C

- **Description:** Read Xref File batch module. Handles mappings of credit card numbers to customer account IDs.
- **Lines of Code:** 145
- **Author:** AWS
- **Logical Files:** XREFFILE-FILE (DD: XREFFILE)
- **Copybooks Referenced:** CVACT03Y
- **Subprograms Called:** CEE3ABD

Program: CBACT04C

- **Description:** Interest Calculation batch module. Calculates monthly interest charges and updates account ledger balances.
- **Lines of Code:** 599
- **Author:** AWS
- **Logical Files:** TCATBAL-FILE (DD: TCATBALF), XREF-FILE (DD: XREFFILE), ACCOUNT-FILE (DD: ACCTFILE), DISCGRP-FILE (DD: DISCGRP), TRANSACT-FILE (DD: TRANSACT)
- **Copybooks Referenced:** CVTRA01Y, CVACT03Y, CVTRA02Y, CVACT01Y, CVTRA05Y
- **Subprograms Called:** CEE3ABD

Program: CBCUS01C

- **Description:** Read Customer File batch module. Extracts details like name, address, and SSN for ingestion.
- **Lines of Code:** 145
- **Author:** AWS
- **Logical Files:** CUSTFILE-FILE (DD: CUSTFILE)
- **Copybooks Referenced:** CVCUS01Y
- **Subprograms Called:** CEE3ABD

Program: CBEXPORT

- **Description:** Data Export batch utility. Backs up VSAM KSDS files into sequential data files.
- **Lines of Code:** 491
- **Author:** CARDDemo TEAM
- **Logical Files:** CUSTOMER-INPUT (DD: CUSTFILE), ACCOUNT-INPUT (DD: ACCTFILE), XREF-INPUT (DD: XREFFILE), TRANSACTION-INPUT (DD: TRANSACT), CARD-INPUT (DD: CARDFILE), EXPORT-OUTPUT (DD: EXPFILE)

- **Copybooks Referenced:** CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT
- **Subprograms Called:** CEE3ABD

Program: CBIMPORT

- **Description:** Data Import batch utility. Restores sequential backups into VSAM KSDS database files.
- **Lines of Code:** 413
- **Author:** CARDDEMO TEAM
- **Logical Files:** EXPORT-INPUT (DD: EXPFILE), CUSTOMER-OUTPUT (DD: CUSTOUT), ACCOUNT-OUTPUT (DD: ACCTOUT), XREF-OUTPUT (DD: XREFOUT), TRANSACTION-OUTPUT (DD: TRNXOUT), CARD-OUTPUT (DD: CARDOUT), ERROR-OUTPUT (DD: ERR0UT)
- **Copybooks Referenced:** CVCUS01Y, CVACT01Y, CVACT03Y, CVTRA05Y, CVACT02Y, CVEXPORT
- **Subprograms Called:** CEE3ABD

Program: CBSTM03A

- **Description:** Statement Generation batch processor. Compiles monthly account details and writes statement summaries.
- **Lines of Code:** 863
- **Author:** AWS
- **Logical Files:** STMT-FILE (DD: STMTFILE), HTML-FILE (DD: HTMLFILE)
- **Copybooks Referenced:** COSTM01, CVACT03Y, CUSTREC, CVACT01Y
- **Subprograms Called:** CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CBSTM03B, CEE3ABD

Program: CBSTM03B

- **Description:** Statement Generation helper sub-routine. Manages PDF rendering details.
- **Lines of Code:** 204
- **Author:** AWS
- **Logical Files:** TRNX-FILE (DD: TRNXFILE), XREF-FILE (DD: XREFFILE), CUST-FILE (DD: CUSTFILE), ACCT-FILE (DD: ACCTFILE)
- **Copybooks Referenced:** None
- **Subprograms Called:** None

Program: CBTRN01C

- **Description:** Daily Transaction Ingestion module. Verifies transactions against active credit cards.
- **Lines of Code:** 449
- **Author:** AWS
- **Logical Files:** DALYTRAN-FILE (DD: DALYTRAN), CUSTOMER-FILE (DD: CUSTFILE), XREF-FILE (DD: XREFFILE), CARD-FILE (DD: CARDFILE), ACCOUNT-FILE (DD: ACCTFILE), TRANSACT-FILE (DD: TRANFILE)
- **Copybooks Referenced:** CVTRA06Y, CVCUS01Y, CVACT03Y, CVACT02Y, CVACT01Y, CVTRA05Y
- **Subprograms Called:** CEE3ABD

Program: CBTRN02C

- **Description:** Daily Transaction Posting batch module. Deducts amounts and updates card accounts balances.
- **Lines of Code:** 672
- **Author:** AWS
- **Logical Files:** DALYTRAN-FILE (DD: DALYTRAN), TRANSACT-FILE (DD: TRANFILE), XREF-FILE (DD: XREFFILE), DALYREJS-FILE (DD: DALYREJS), ACCOUNT-FILE (DD: ACCTFILE), TCATBAL-FILE (DD: TCATBALF)
- **Copybooks Referenced:** CVTRA06Y, CVTRA05Y, CVACT03Y, CVACT01Y, CVTRA01Y
- **Subprograms Called:** CEE3ABD

Program: CBTRN03C

- **Description:** Transaction Report generator batch utility. Reports daily totals categorized by transaction types.
- **Lines of Code:** 596
- **Author:** AWS
- **Logical Files:** TRANSACT-FILE (DD: TRANFILE), XREF-FILE (DD: CARDXREF), TRANTYPE-FILE (DD: TRANTYPE), TRANCATG-FILE (DD: TRANCATG), REPORT-FILE (DD: TRANREPT), DATE-PARMS-FILE (DD: DATEPARM)
- **Copybooks Referenced:** CVTRA05Y, CVACT03Y, CVTRA03Y, CVTRA04Y, CVTRA07Y
- **Subprograms Called:** CEE3ABD

Program: COACTUPC

- **Description:** CICS Online Account Update screen controller. Updates customer contact profiles.
- **Lines of Code:** 3744

- [illegible]

Program: COACTVWC

- **Description:** CICS Online Account View screen controller. Searches and displays customer account summaries.
- **Lines of Code:** 811
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COACTVW, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT01Y, CVACT02Y, CVACT03Y, CVCUS01Y
- **Subprograms Called:** None
- **IDMS Records:** ASSOCIATED, THIS

Program: COADM01C

- **Description:** CICS Online Admin Menu screen controller. Routes to admin tools.
- **Lines of Code:** 230
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COADM02Y, COADM01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COBIL00C

- **Description:** CICS Online Bill Payment screen controller. Logs payments and checks balances.
- **Lines of Code:** 493

- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COBIL00, COTT01Y, CSDAT01Y, CSMSG01Y, CVACT01Y, CVACT03Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COBSWAIT

- **Description:** Wait utility module. Uses Assembler subroutine MVSWAIT to pause step executions.
- **Lines of Code:** 19
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** None
- **Subprograms Called:** MVSWAIT

Program: COCRDLIC

- **Description:** CICS Online Credit Card List screen controller. Lists credit card details.
- **Lines of Code:** 1256
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COCRDLI, CSDAT01Y, CSMSG01Y, CSUSR01Y, CVACT02Y
- **Subprograms Called:** None

Program: COCRDSL

- **Description:** CICS Online Credit Card Detail screen controller. Displays credit card balances and interest rates.
- **Lines of Code:** 757
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COCRDSL, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y
- **Subprograms Called:** None
- **IDMS Records:** CARDS, THIS

Program: COCRDUPC

- **Description:** CICS Online Credit Card Update screen controller. Modifies credit card limits and expiration dates.
- **Lines of Code:** 1365
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** CVCRD01Y, COCOM01Y, DFHBMSCA, DFHAID, COTT01Y, COCRDUP, CSDAT01Y, CSMSG01Y, CSMSG02Y, CSUSR01Y, CVACT02Y, CVCUS01Y
- **Subprograms Called:** None
- **IDMS Records:** CARDS, THIS

Program: COMEN01C

- **Description:** CICS Online Main Menu controller. Main navigation hub routing CICS terminal sessions.
- **Lines of Code:** 255
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COMEN02Y, COMEN01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: CORPT00C

- **Description:** CICS Online Report viewer. Displays calculated statements on CICS terminal panels.
- **Lines of Code:** 586
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, CORPT00, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** CSUTLDT, CSUTLDT

Program: COSGN00C

- **Description:** CICS Online Signon screen controller. Performs terminal user authentication and security checks.
- **Lines of Code:** 211

- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COSGN00, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COTRN00C

- **Description:** CICS Online Transaction Menu screen controller. Lists transaction histories.
- **Lines of Code:** 617
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COTRN00, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COTRN01C

- **Description:** CICS Online Transaction Detail screen controller. Displays details of a specific transaction.
- **Lines of Code:** 273
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COTRN01, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COTRN02C

- **Description:** CICS Online Transaction Add screen controller. Enters new transaction entries.
- **Lines of Code:** 698
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COTRN02, COTT01Y, CSDAT01Y, CSMSG01Y, CVTRA05Y, CVACT01Y, CVACT03Y, DFHAID, DFHBMSCA
- **Subprograms Called:** CSUTLDTC, CSUTLDTC

Program: COUSR00C

- **Description:** CICS Online User Control screen controller. Inventories CICS account profile logins.
- **Lines of Code:** 615
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR00, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COUSR01C

- **Description:** CICS Online User Add screen controller. Adds new user credentials.
- **Lines of Code:** 238
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR01, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None

Program: COUSR02C

- **Description:** CICS Online User Edit screen controller. Updates user roles.
- **Lines of Code:** 351
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR02, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA
- **Subprograms Called:** None
- **IDMS Records:** T0

Program: COUSR03C

- **Description:** CICS Online User Delete screen controller. Removes CICS credentials.
- **Lines of Code:** 296
- **Author:** AWS
- **Logical Files:** None
- **Copybooks Referenced:** COCOM01Y, COUSR03, COTT01Y, CSDAT01Y, CSMSG01Y, CSUSR01Y, DFHAID, DFHBMSCA

- **Subprograms Called:** None

Program: CSUTLDTG

- **Description:** Date Converter utility sub-routine. Standardizes various date formats.
- **Lines of Code:** 128
- **Author:** UNKNOWN
- **Logical Files:** None
- **Copybooks Referenced:** None
- **Subprograms Called:** CEEDAYS

4.5 JCL Job Stream Analysis

Detailed operational breakdown of JCL jobs schedules, step sequences, and execution configurations:

JCL Job: ACCTFILE

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), ACCTDATA (DSN: AWS.M2.CARDDemo.ACCTDATA.PS, DISP: SHR), ACCTVSAM (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: CARDFILE

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (8):**

Step CLCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), CARDDATA (DSN: AWS.M2.CARDDemo.CARDDATA.PS, DISP: SHR), CARDVSAM (DSN: AWS.M2.CARDDemo.CARDDATA.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP40: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP50: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP60: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step OPCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

JCL Job: CBADMCDJ

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP1: Executes program DFHCSDUP.

- DD Assignments: STEPLIB (DSN: OEM.CICSTS.V05R06M0.CICS.SDFHLOAD, DISP: SHR), DFHCSD (DSN: OEM.CICSTS.DFHCSD, DISP: SHR), OUTDD (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: CBEXPORT

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (2):**

Step STEP01: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP02: Executes program CEXPORT.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), CUSTFILE (DSN: AWS.M2.CARDDemo.CUSTDATA.VSAM.KSDS, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), TRANSACT (DSN: AWS.M2.CARDDemo.TRANSACT.VSAM.KSDS, DISP: SHR), CARDFILE (DSN: AWS.M2.CARDDemo.CARDDATA.VSAM.KSDS, DISP: SHR), EXPFILE (DSN: AWS.M2.CARDDemo.EXPORT.DATA, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR)

JCL Job: CBIMPORT

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP01: Executes program CBIMPORT.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), EXPFILE (DSN: AWS.M2.CARDDemo.EXPORT.DATA, DISP: SHR), CUSTOUT (DSN: AWS.M2.CARDDemo.CUSTDATA.IMPORT, DISP: NEW, CATLG, DELETE), ACCTOUT (DSN: AWS.M2.CARDDemo.ACCTDATA.IMPORT, DISP: NEW, CATLG, DELETE), XREFOUT (DSN: AWS.M2.CARDDemo.CARDXREF.IMPORT, DISP: NEW, CATLG, DELETE), TRNXOUT (DSN: AWS.M2.CARDDemo.TRANSACT.IMPORT, DISP: NEW, CATLG, DELETE), ERRORT (DSN: AWS.M2.CARDDemo.IMPORT.ERRORS, DISP: NEW, CATLG, DELETE), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR)

JCL Job: CLOSEFIL

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step CLCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

JCL Job: COMBTRAN

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (2):**

Step STEP05R: Executes program SORT.

- DD Assignments: SORTIN (DSN: AWS.M2.CARDDemo.TRANSACT.BKUP(0), DISP: SHR), SYMNames (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDemo.TRANSACT.COMBINED(, DISP: NEW, CATLG, DELETE)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), TRANSACT (DSN: AWS.M2.CARDDemo.TRANSACT.COMBINED(, DISP: SHR), TRANVSAM (DSN: AWS.M2.CARDDemo.TRANSACT.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: CRESTMT

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (5):**

Step DELDEF01: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP010: Executes program SORT.

- DD Assignments: SORTIN (DSN: AWS.M2.CARDDemo.TRANSACT.VSAM.KSDS, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDemo.TRXFL.SEQ, DISP: NEW,CATLG,DELETE), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP020: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), INFILE (DSN: AWS.M2.CARDDemo.TRXFL.SEQ, DISP: SHR), OUTFILE (DSN: AWS.M2.CARDDemo.TRXFL.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP030: Executes program IEFBR14.

- DD Assignments: HTMLFILE (DSN: AWS.M2.CARDDemo.STATEMNT.HTML, DISP: MOD,DELETE,DELETE), STMTFILE (DSN: AWS.M2.CARDDemo.STATEMNT.PS, DISP: MOD,DELETE,DELETE)

Step STEP040: Executes program CBSTM03A.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), TRNXFILE (DSN: AWS.M2.CARDDemo.TRXFL.VSAM.KSDS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), CUSTFILE (DSN: AWS.M2.CARDDemo.CUSTDATA.VSAM.KSDS, DISP: SHR), STMTFILE (DSN: SYSOUT, DISP: NEW,CATLG,DELETE), HTMLFILE (DSN: AWS.M2.CARDDemo.STATEMNT.HTML, DISP: NEW,CATLG,DELETE)

JCL Job: CUSTFILE

- **Job Stream Description:** Sequential execution of batch components.

- **Steps Defined (5):**

Step CLCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), CUSTDATA (DSN: AWS.M2.CARDDemo.CUSTDATA.PS, DISP: SHR), CUSTVSAM (DSN: AWS.M2.CARDDemo.CUSTDATA.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step OPCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

JCL Job: DALYREJS

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: DEFCUST

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (2):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: DEFGDGB

- **Job Stream Description:** Sequential execution of batch components.

- **Steps Defined (1):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: DEFGDGD

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (6):**

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP20: Executes program IEBGENER.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR), SYSUT1 (DSN: AWS.M2.CARDDemo.TRANTYPE.PS, DISP: SHR), SYSUT2 (DSN: AWS.M2.CARDDemo.TRANTYPE.BKUP(, DISP: NEW, CATLG)

Step STEP30: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP40: Executes program IEBGENER.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR), SYSUT1 (DSN: AWS.M2.CARDDemo.TRANCATG.PS, DISP: SHR), SYSUT2 (DSN: AWS.M2.CARDDemo.TRANCATG.PS.BKUP(, DISP: NEW, CATLG)

Step STEP50: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP60: Executes program IEBGENER.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR), SYSUT1 (DSN: AWS.M2.CARDDemo.DISCGRP.PS, DISP: SHR), SYSUT2 (DSN: AWS.M2.CARDDemo.DISCGRP.BKUP(, DISP: NEW, CATLG)

JCL Job: DISCGRP

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), DISCGRP (DSN: AWS.M2.CARDDemo.DISCGRP.PS, DISP: SHR), DISCVSAM (DSN: AWS.M2.CARDDemo.DISCGRP.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: DUSRSECJ

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (4):**

Step PREDEL: Executes program IEFBR14.

- DD Assignments: DD01 (DSN: AWS.M2.CARDDemo.USRSEC.PS, DISP: MOD, DELETE, DELETE)

Step STEP01: Executes program IEBGENER.

- DD Assignments: SYSUT1 (DSN: SYSOUT, DISP: SHR), SYSUT2 (DSN: AWS.M2.CARDDemo.USRSEC.PS, DISP: NEW, CATLG, DELETE), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP02: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP03: Executes program IDCAMS.

- DD Assignments: IN (DSN: AWS.M2.CARDDemo.USRSEC.PS, DISP: SHR), OUT (DSN: AWS.M2.CARDDemo.USRSEC.VSAM.KSDS, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: ESDSRRDS

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (6):**

Step PREDEL: Executes program IEFBR14.

- DD Assignments: DD01 (DSN: AWS.M2.CARDDemo.ESDSRRDS.PS, DISP: MOD, DELETE, DELETE)

Step STEP01: Executes program IEBGENER.

- DD Assignments: SYSUT1 (DSN: SYSOUT, DISP: SHR), SYSUT2 (DSN: AWS.M2.CARDDemo.ESDSRRDS.PS, DISP: NEW, CATLG, DELETE), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP02: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP03: Executes program IDCAMS.

- DD Assignments: IN (DSN: AWS.M2.CARDDemo.ESDSRRDS.PS, DISP: SHR), OUT (DSN: AWS.M2.CARDDemo.USRSEC.VSAM.ESDS, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP04: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP05: Executes program IDCAMS.

- DD Assignments: IN (DSN: AWS.M2.CARDDemo.ESDSRRDS.PS, DISP: SHR), OUT (DSN: AWS.M2.CARDDemo.USRSEC.VSAM.RRDS, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: FTPJCLS

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP1: Executes program FTP.

- DD Assignments: SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: INTCALC

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP15: Executes program CBACT04C.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), TCATBALF (DSN: AWS.M2.CARDDemo.TCATBALF.VSAM.KSDS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), XREFFIL1 (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.AIX.PATH, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), DISCGRP (DSN: AWS.M2.CARDDemo.DISCGRP.VSAM.KSDS, DISP: SHR), TRANSACT (DSN: AWS.M2.CARDDemo.SYSTRAN(, DISP: NEW, CATLG, DELETE)

JCL Job: INTRDRJ1

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (2):**

Step IDCAMS: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), IN (DSN: AWS.M2.CARDEMO.FTP.TEST, DISP: SHR), OUT (DSN: AWS.M2.CARDEMO.FTP.TEST.BKUP, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP01: Executes program IEBGENER.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR), SYSUT1 (DSN: AWS.M2.CARDDemo.JCL (INTRDRJ2), DISP: SHR), SYSUT2 (DSN: SYSOUT, DISP: SHR)

JCL Job: INTRDRJ2

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step IDCAMS: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), IN (DSN: AWS.M2.CARDEMO.FTP.TEST.BKUP, DISP: SHR), OUT (DSN: AWS.M2.CARDEMO.FTP.TEST.BKUP.INTRDR, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: OPENFIL

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step OPCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

JCL Job: POSTTRAN

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP15: Executes program CBTRN02C.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), TRANFILE (DSN: AWS.M2.CARDDemo.TRANSACTION.VSAM.KSDS, DISP: SHR), DALYTRAN (DSN: AWS.M2.CARDDemo.DALYTRAN.PS, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), DALYREJS (DSN: AWS.M2.CARDDemo.DALYREJS, DISP: NEW, CATLG, DELETE), ACCTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), TCATBALF (DSN: AWS.M2.CARDDemo.TCATBALF.VSAM.KSDS, DISP: SHR)

JCL Job: PRTCATBL

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**

Step DELDEF: Executes program IEFBR14.

- DD Assignments: THEFILE (DSN: AWS.M2.CARDDemo.TCATBALF.REPT, DISP: MOD, DELETE)

- **Step STEP05R:** Executes program REPROC.

Step STEP10R: Executes program SORT.

- DD Assignments: SORTIN (DSN: AWS.M2.CARDDemo.TCATBALF.BKUP(, DISP: SHR), SYMNames (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDemo.TCATBALF.REPT, DISP: NEW, CATLG, DELETE)

JCL Job: READACCT

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (2):**

Step PREDEL: Executes program IEFBR14.

- DD Assignments: DD01 (DSN: AWS.M2.CARDDemo.ACCTDATA.PSCOMP, DISP: MOD, DELETE, DELETE), DD02 (DSN: AWS.M2.CARDDemo.ACCTDATA.ARRYPS, DISP: MOD, DELETE, DELETE), DD03 (DSN: AWS.M2.CARDDemo.ACCTDATA.VBPS, DISP: MOD, DELETE, DELETE)

Step STEP05: Executes program CBACT01C.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), ACCTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VSAM.KSDS, DISP: SHR), OUTFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.PSCOMP, DISP: NEW, CATLG, DELETE), ARRYFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.ARRYPS, DISP: NEW, CATLG, DELETE), VBRCFILE (DSN: AWS.M2.CARDDemo.ACCTDATA.VBPS, DISP: NEW, CATLG, DELETE), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR)

JCL Job: READCARD

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP05: Executes program CBACT02C.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), CARDFILE (DSN: AWS.M2.CARDDemo.CARDDATA.VSAM.KSDS, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR)

JCL Job: READCUST

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP05: Executes program CBCUS01C.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), CUSTFILE (DSN: AWS.M2.CARDDemo.CUSTDATA.VSAM.KSDS, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR)

JCL Job: READXREF

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP05: Executes program CBACT03C.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), XREFFILE (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR)

JCL Job: REPTFILE

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: TCATBALF

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), TCATBAL (DSN: AWS.M2.CARDDemo.TCATBALF.PS, DISP: SHR), TCATBALV (DSN: AWS.M2.CARDDemo.TCATBALF.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: TRANBKP

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**
- **Step STEP05R:** Executes program REPROC.

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: TRANCATG

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), TRANCATG (DSN: AWS.M2.CARDDemo.TRANCATG.PS, DISP: SHR), TCATVSAM (DSN: AWS.M2.CARDDemo.TRANCATG.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: TRANFILE

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (8):**

Step CLCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), TRANSACT (DSN: AWS.M2.CARDDemo.DALYTRAN.PS.INIT, DISP: SHR), TRANVSAM (DSN: AWS.M2.CARDDemo.TRANSACTION.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP20: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP25: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP30: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step OPCIFIL: Executes program SDSF.

- DD Assignments: ISFOUT (DSN: SYSOUT, DISP: SHR), CMDOUT (DSN: SYSOUT, DISP: SHR), ISFIN (DSN: SYSOUT, DISP: SHR)

JCL Job: TRANIDX

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**

Step STEP20: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP25: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP30: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: TRANREPT

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**
- **Step STEP05R:** Executes program REPROC.

Step STEP05R: Executes program SORT.

- DD Assignments: SORTIN (DSN: AWS.M2.CARDDemo.TRANSACTION.BKUP(, DISP: SHR), SYMNames (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR), SYSOUT (DSN:

SYSOUT, DISP: SHR), SORTOUT (DSN: AWS.M2.CARDDemo.TRANSACTION.DALY(, DISP: NEW, CATLG, DELETE)

Step STEP10R: Executes program CBTRN03C.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), TRANFILE (DSN: AWS.M2.CARDDemo.TRANSACTION.DALY(, DISP: SHR), CARDXREF (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), TRANTYPE (DSN: AWS.M2.CARDDemo.TRANTYPE.VSAM.KSDS, DISP: SHR), TRANCATG (DSN: AWS.M2.CARDDemo.TRANCATG.VSAM.KSDS, DISP: SHR), DATEPARM (DSN: AWS.M2.CARDDemo.DATEPARM, DISP: SHR), TRANREPT (DSN: AWS.M2.CARDDemo.TRANREPT(, DISP: NEW, CATLG, DELETE)

JCL Job: TRANTYPE

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (3):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), TRANTYPE (DSN: AWS.M2.CARDDemo.TRANTYPE.PS, DISP: SHR), TTYPSAM (DSN: AWS.M2.CARDDemo.TRANTYPE.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: TXT2PDF1

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (1):**

Step TXT2PDF: Executes program IKJEFT1B.

- DD Assignments: STEPLIB (DSN: AWS.M2.LBD.TXT2PDF.LOAD, DISP: SHR), SYSEXEC (DSN: AWS.M2.LBD.TXT2PDF.EXEC, DISP: SHR), INDD (DSN: AWS.M2.CARDDemo.STATEMNT.PS, DISP: SHR), SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSTSPRT (DSN: SYSOUT, DISP: SHR), SYSTSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: WAITSTEP

- **Job Stream Description:** Sequential execution of batch components.

- **Steps Defined (1):**

Step WAIT: Executes program COBSWAIT.

- DD Assignments: STEPLIB (DSN: AWS.M2.CARDDemo.LOADLIB, DISP: SHR), SYSOUT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

JCL Job: XREFFILE

- **Job Stream Description:** Sequential execution of batch components.
- **Steps Defined (6):**

Step STEP05: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP10: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP15: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), XREFDATA (DSN: AWS.M2.CARDDemo.CARDXREF.PS, DISP: SHR), XREFVSAM (DSN: AWS.M2.CARDDemo.CARDXREF.VSAM.KSDS, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP20: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP25: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

Step STEP30: Executes program IDCAMS.

- DD Assignments: SYSPRINT (DSN: SYSOUT, DISP: SHR), SYSIN (DSN: SYSOUT, DISP: SHR)

4.6 Assembler Module Analysis

Lower-level HLASM modules referenced by subprograms in execution flows:

Assembler Module: COBDATFT

- **Description:** Assembler date formatting subroutine.
- **Entry Points:** COBDATFT
- **Calls Made:** None

Assembler Module: MVSWAIT

- **Description:** Assembler interval wait subroutine.
- **Entry Points:**
- **Calls Made:** None

4.7 CICS BMS Mapset Layouts

Defined CICS terminal screens, field layouts, and positioning mappings:

Mapset: COACTUP

- **Maps Included:** CACTUPA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACSTTUS (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), OPNYEAR (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 1)

Mapset: COACTVW

- **Maps Included:** CACTVWA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACSTTUS (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ADT0PEN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COADM01

- **Maps Included:** COADM1A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), OPTN001 (Pos: N/A, Len: 0), OPTN002 (Pos: N/A, Len: 0), OPTN003 (Pos: N/A, Len: 0), OPTN004 (Pos: N/A, Len: 0), OPTN005 (Pos: N/A, Len: 0), OPTN006 (Pos: N/A, Len: 0), OPTN007 (Pos: N/A, Len: 0), OPTN008 (Pos: N/A, Len: 0), OPTN009 (Pos: N/A, Len: 0)

Mapset: COBIL00

- **Maps Included:** COBIL0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACTIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURBAL (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CONFIRM (Pos: N/A, Len: 0)

Mapset: COCRDLI

- **Maps Included:** CCRDLIA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 5), PAGENO (Pos: N/A, Len: 3), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COCRDSL

- **Maps Included:** CCRDSLA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CRDNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COCRDUP

- **Maps Included:** CCRDUPA
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACCTSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CRDNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COMEN01

- **Maps Included:** COMEN1A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), OPTN001 (Pos: N/A, Len: 0), OPTN002 (Pos: N/A, Len: 0), OPTN003 (Pos: N/A, Len: 0), OPTN004 (Pos: N/A, Len: 0), OPTN005 (Pos: N/A, Len: 0), OPTN006 (Pos: N/A, Len: 0), OPTN007 (Pos: N/A, Len: 0), OPTN008 (Pos: N/A, Len: 0), OPTN009 (Pos: N/A, Len: 0)

Mapset: CORPT00

- **Maps Included:** CORPT0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), MONTHLY (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), YEARLY (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CUSTOM (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COSGN00

- **Maps Included:** COSGN0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), APPLID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), SYSID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COTRN00

- **Maps Included:** COTRN0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PAGENUM (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), TRNIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COTRN01

- **Maps Included:** COTRN1A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), TRNIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), TRNID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDNUM (Pos: N/A, Len: 0)

Mapset: COTRN02

- **Maps Included:** COTRN2A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), ACTIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CARDNIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COUSR00

- **Maps Included:** COUSR0A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PAGENUM (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USRIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COUSR01

- **Maps Included:** COUSR1A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), LNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USERID (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0)

Mapset: COUSR02

- **Maps Included:** COUSR2A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USRIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), LNAME (Pos: N/A, Len: 0)

Mapset: COUSR03

- **Maps Included:** COUSR3A
- **Field Layouts (Top 20):** FILLER (Pos: N/A, Len: 0), TRNNAME (Pos: N/A, Len: 0), TITLE01 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURDATE (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), PGMNAME (Pos: N/A, Len: 0), TITLE02 (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), CURTIME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), USRIDIN (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FNAME (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), FILLER (Pos: N/A, Len: 0), LNAME (Pos: N/A, Len: 0)